The Satellite1 Ultra Build Book

DO NOT PRINT THE FULL ENCLOSURE YET.
PRINT AND MEASURE THE 8 SMALL CALIBRATION PIECES FIRST.

This is the only document you need to follow from start to finish. Work through the phases in order. Every other guide in this package is reference material that this book points you to at the moment you need it.

You are building a serviceable passive-radiator speaker enclosure around the FutureProofHomes Satellite1 **Batch 1** development kit: Core rev4.1 and HAT rev4.1 / R2024.12.06. Satellite1.1 / Batch 2 does not fit and is not supported.

Expected difficulty is 4 out of 5. Nothing here is glued; every joint is a screw into a brass insert, so you can open the unit again later.

The finished Satellite1 Ultra, assembled

The whole build at a glance

Phase	What you do	Roughly how long
1	Check your printer and buy the parts	ordering time
2	Print 8 small calibration pieces	4-6 hours
3	Measure them and correct the files	1 hour
4	Print 12 Ultra parts and 6 Satellite top parts	3-5 days
5	Cut 3 foam gaskets from the templates	30 minutes
6	Lay out and check every part	30 minutes
7	Assemble, in 9 numbered steps	3-4 hours
8	Test before you use it normally	2 hours

Status you should know before you start

Every gate in this repository is a **digital** check: geometry, clearances, volumes, exports, and documents. No physical unit has been built and measured. Fit, sealing, acoustic output, thermals, Wi-Fi, microphones, buttons, LEDs, and wake-word behaviour are all `REQUIRES_PHYSICAL_VALIDATION`. Your calibration, leak, and commissioning checks in phases 3 and 8 are part of the build, not optional extras.

Phase 1 — Check your printer and buy the parts

Your printer must actually fit the shell

The outer shell is the limiting part at **192 x 212 x 189 mm**, printed upright. You therefore need **212 x 192 x 189 mm of genuinely usable travel** (X and Y may be swapped).

• A 220 x 220 x 200 mm printer works only if the whole bed is usable. That

leaves about 4 mm per long-side edge and limits the shell brim to 3 mm.

• Purge lines, bed clips, and firmware exclusion zones all steal usable travel.

Measure what your machine really reaches before you commit.

• Rotating the shell in-plane does not make it fit a 210 x 210 mm bed, and

printing it on its side is not supported.

You also need an enclosed printer, a 0.4 mm nozzle, and dry filament.

Materials

• **ASA** for every rigid part. **PETG** is a documented alternative.

• **TPU 95A** for the two flexible parts: the cable seal and the bottom grip.

• **PLA+** is only acceptable for a display mock-up, never a finished unit.

Tools

Digital calipers reading 0.01 mm, a scale reading 0.01 g, a 2.0 mm hex driver, a temperature-controlled soldering iron with an M3 heat-set-insert tip, wire strippers and a crimper or soldering iron, ESD protection, a hand bulb with a 0-500 Pa gauge, and leak-detection solution.

What to buy

Buy every item marked required in `BOM.csv`. The essentials are:

• One FutureProofHomes Satellite1 Batch 1 kit (E01).

• One Dayton Audio ND91-4 speaker (A01).

• Two SB Acoustics SB12PACR-00 passive radiators (A02).

• M3 heat-set inserts, with four spares (H01).

• Every M3 screw in `FASTENERS.csv`.

• One 300 x 300 mm sheet of 2.0 mm closed-cell EPDM foam (G00).

• Two mild-steel ballast plates (B01).

• Two matched M6 washer stacks for radiator tuning (B02).

• One 2-pin JST-XH speaker lead (H02).

`HARDWARE_AND_MATERIALS_GUIDE.pdf` gives the full specification for each line.

Do not print the official original speaker chamber, speaker plate, or
anti-slip ring. The Ultra parts replace all three.

Phase 2 — Print the 8 calibration pieces

These are small. They exist so you discover your printer's real dimensional behaviour before you spend days of filament on the big parts.

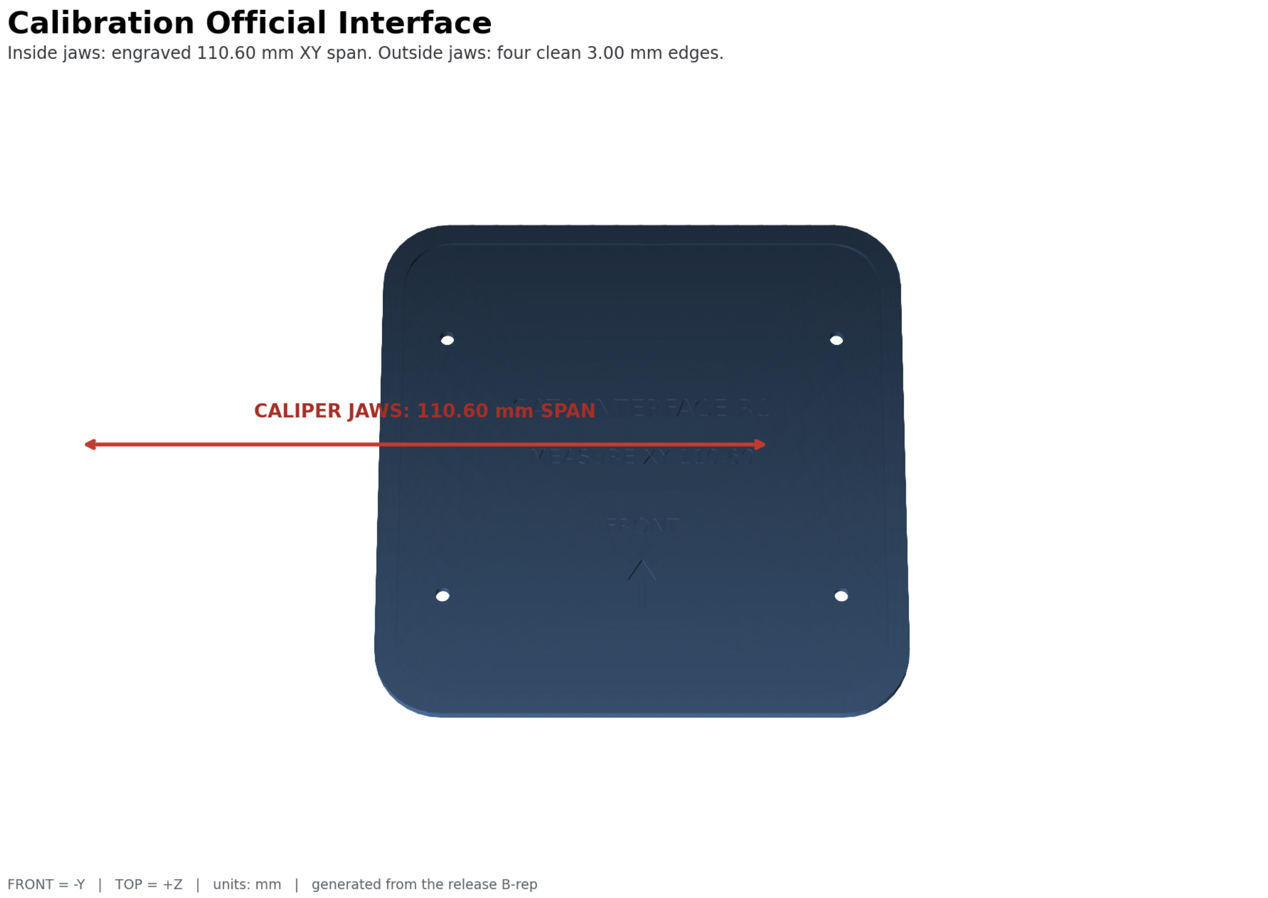
Open `PRINT_THESE_FILES/1_CALIBRATION_FIRST` and print every file, one at a time, using the exact settings you will use for the enclosure:

File	Print this many	Material
01_CHECK_SATELLITE_TOP_FIT.3mf	1	ASA
02_CHECK_SCREWS_AND_INSERTS.3mf	1	ASA
03_CHECK_SPEAKER_FIT.3mf	1	ASA
04_CHECK_RADIATOR_FIT.3mf	1	ASA
05_GASKET_TEST_BASE.3mf	1	ASA
06_GASKET_TEST_TOP.3mf	1	ASA
07_CHECK_CABLE_HOLE.3mf	1	ASA
08_FLEXIBLE_CABLE_SEAL_TPU.3mf	1	TPU 95A

Baseline settings: 0.4 mm nozzle, 0.20 mm layers, five walls, six top and six bottom layers, 35% gyroid infill, no supports, 5 mm brim on the rigid pieces. ASA runs 250-260 C nozzle and 100-110 C bed with the enclosure closed. PETG runs 235-250 C nozzle and 75-85 C bed.

Phase 3 — Measure the pieces and correct the files

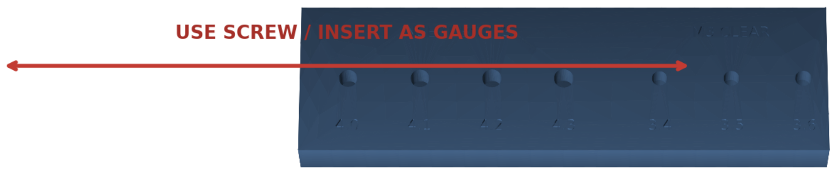
This is the step that makes the big parts fit. `CALIBRATION_GUIDE.pdf` shows exactly where each caliper jaw goes; the images below name each check.



Where to measure the Satellite top fit piece

Calibration Fasteners

Functional gauges: M3 screw in 3.4/3.5/3.6 holes; insert in 4.0–4.3 blind bores.



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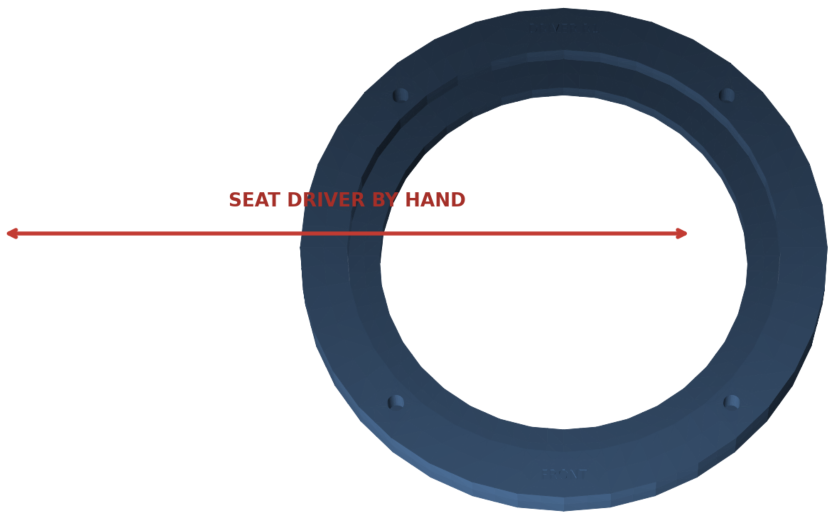
Using the screw and insert as go/no-go gauges

Calibration Driver

Seat the purchased ND91-4 by hand. Measure its flange thickness at four quadrants.



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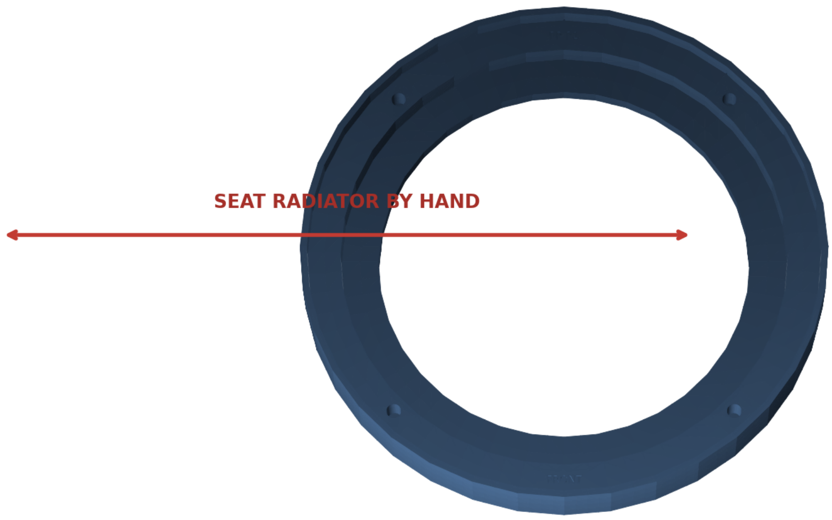
Checking the speaker seats by hand

Calibration Radiator

Seat one SB12PACR-00 by hand. Measure its flange thickness at four quadrants.



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Checking a radiator seats by hand

Calibration Gasket

Measure sheet thickness; tighten until both hard stops touch; inspect the closed light path.

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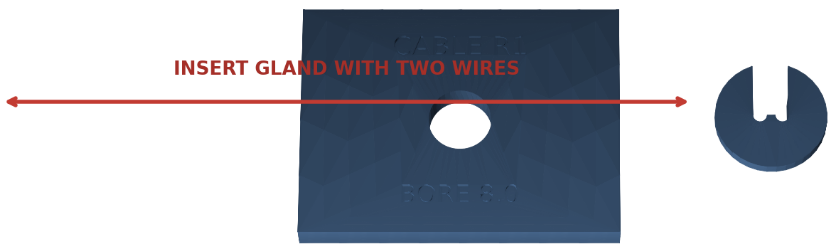
Squeezing the foam between the two gasket pieces

Calibration Cable

Fit the two actual 22 AWG conductors. The gland must seat by hand and resist rotation.



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Fitting the two real wires through the cable seal

The eight checks

Check	What you do	It passes when
XY scale	Inside caliper jaws across the engraved 110.60 mm recess, at three heights	110.40-110.80 mm after correction
Z scale	Outside jaws across a clean 3.00 mm edge, at four corners	2.90-3.10 mm
Screw clearance	Try a clean M3 screw in the 3.4, 3.5 and 3.6 mm holes	smallest hole the screw falls through without force
Insert bore	Install identical inserts into the 4.0-4.3 mm blind bores	square and flush, no crack, no spin
Speaker fit	Seat the real ND91-4 in its test ring	drops in by hand, flange lies flat, play under 0.30 mm
Radiator fit	Seat one real SB12PACR-00 in its test ring	drops in by hand, flange lies flat, play under 0.30 mm
Gasket squeeze	Clamp a strip of your actual foam until both hard stops touch	15%-45% compression, no light path, foam not cut
Cable seal	Push the two real conductors and the TPU seal into the test hole	moderate finger force, seal cannot rotate or lift

Do not measure a 3-4 mm hole with caliper tips. The screw and the insert are the gauges.

Enter your numbers

You have three ways to turn those measurements into corrected files. All three produce the same `physical_calibration.yaml`.

1. **In your browser (easiest).** Open the calibration wizard page listed in

the project README, type in what you measured, and it computes and downloads the file for you. Nothing to install.

2. **On GitHub.** Paste the file contents into the "Calibrated build" workflow

and GitHub rebuilds the whole corrected package for you to download.

3. **On your own computer.** Run `python scripts/calibrate.py`, which asks the

same questions and then runs `make calibrated-release`.

Reprint every calibration piece affected by a correction you entered, and re-check it. Continue only when all eight checks pass on corrected pieces.

If a piece fails, do not sand it and do not "make it work". Correct the
number, regenerate, and reprint. Sanding a sealing face destroys the seal.

Phase 4 — Print every enclosure part

Now print the real parts. Open `PRINT_THESE_FILES/2_ULTRA_ENCLOSURE_PARTS` and print every file:

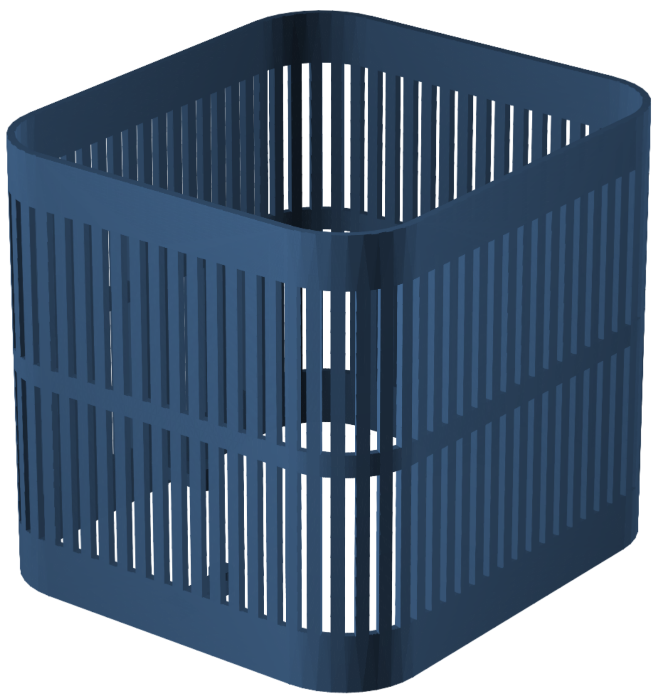
File	Print this many	Material
01_MAIN_SPEAKER_BODY.3mf	1	ASA
02_ELECTRONICS_DIVIDER.3mf	1	ASA

File	Print this many	Material
03_SPEAKER_CLAMP_RING.3mf	1	ASA
04_RADIATOR_CLAMP_RING_PRINT_TWO.3mf	2	ASA
05_BOTTOM_BASE.3mf	1	ASA
06_WEIGHT_TRAY.3mf	1	ASA
07_WEIGHT_TRAY_LID.3mf	1	ASA
08_BOTTOM_ACCESS_PANEL.3mf	1	ASA
09_ELECTRONICS_COVER.3mf	1	ASA
10_OUTER_SHELL.3mf	1	ASA
11_FLEXIBLE_BOTTOM_GRIP_TPU.3mf	1	TPU 95A
12_LEAK_TEST_TOOL.3mf	1	ASA

Note that the radiator clamp ring is the only part you print **twice**.

Print orientation — outer_shell

BED = Z=0. upright, base band on the bed. Keep sealing and insert faces free of support scars.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

The outer shell, positioned on the bed

Then open `PRINT_THESE_FILES/3_SQUIRCLE_TOP_PARTS` and print all six official Satellite top parts. These are not optional; they complete the normal Satellite top:

File	Print this many	Material
01_SATELLITE_MID_PLATE.stl	1	ASA
02_SATELLITE_THREADED_PLATE.stl	1	ASA
03_CIRCUIT_BOARD_SPACER.stl	1	ASA
04_TOP_LOCK_RING.stl	1	ASA
05_BUTTON_AND_LIGHT_TOP.stl	1	ASA

File	Print this many	Material
06_SNAP_IN_LIGHT_RING.stl	1	ASA

`PRINTING_GUIDE.pdf` holds the full slicer baseline, the per-part orientation sheets, and the inspection checklist. Check each part as it comes off the bed: gasket lands must be continuous and unbroken, shell slots must all be open, clamp rings must be flat within 0.20 mm, and no insert bore may have opened into the chamber.

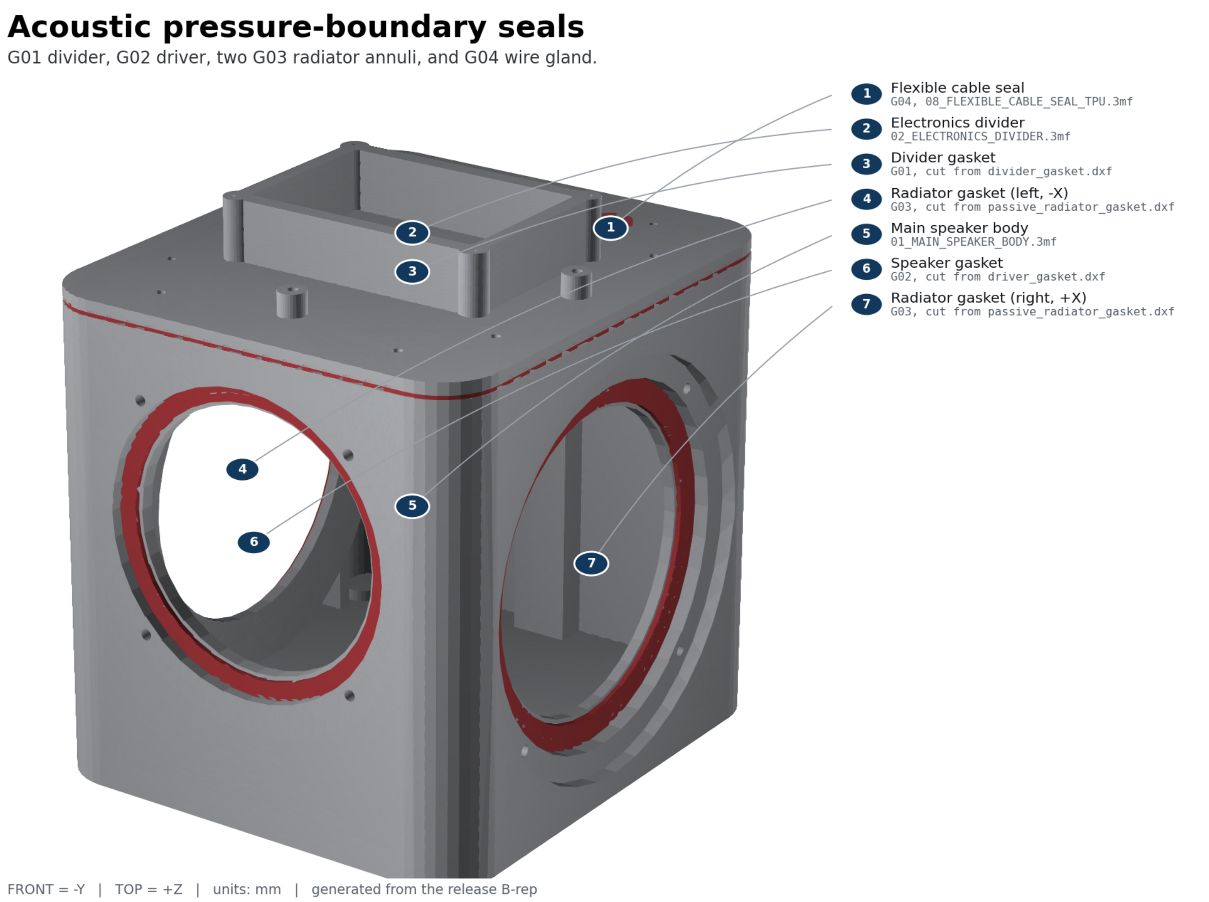
Phase 5 — Cut the three foam gaskets

The gaskets are not printed. You cut them from your 2.0 mm EPDM sheet using the 1:1 templates in `GASKET_TEMPLATES/`. Print each template at exactly 100% scale (no "fit to page"), check the printed size against the sheet, then cut with a sharp knife or a cutting plotter.

ID	Seal	How many	Material	Template
G01	divider_gasket	1	2.0 mm closed-cell EPDM, soft, smooth skin	GASKET_TEMPLATES/divider_gasket.dxf
G02	driver_gasket	1	2.0 mm closed-cell EPDM, soft, smooth skin	GASKET_TEMPLATES/driver_gasket.dxf
G03	passive_radiator_gasket	2	2.0 mm closed-cell EPDM, soft, smooth skin	GASKET_TEMPLATES/passive_radiator_gasket.dxf

The fourth seal, G04, is the flexible cable seal you already printed in TPU as `08_FLEXIBLE_CABLE_SEAL_TPU.3mf`.

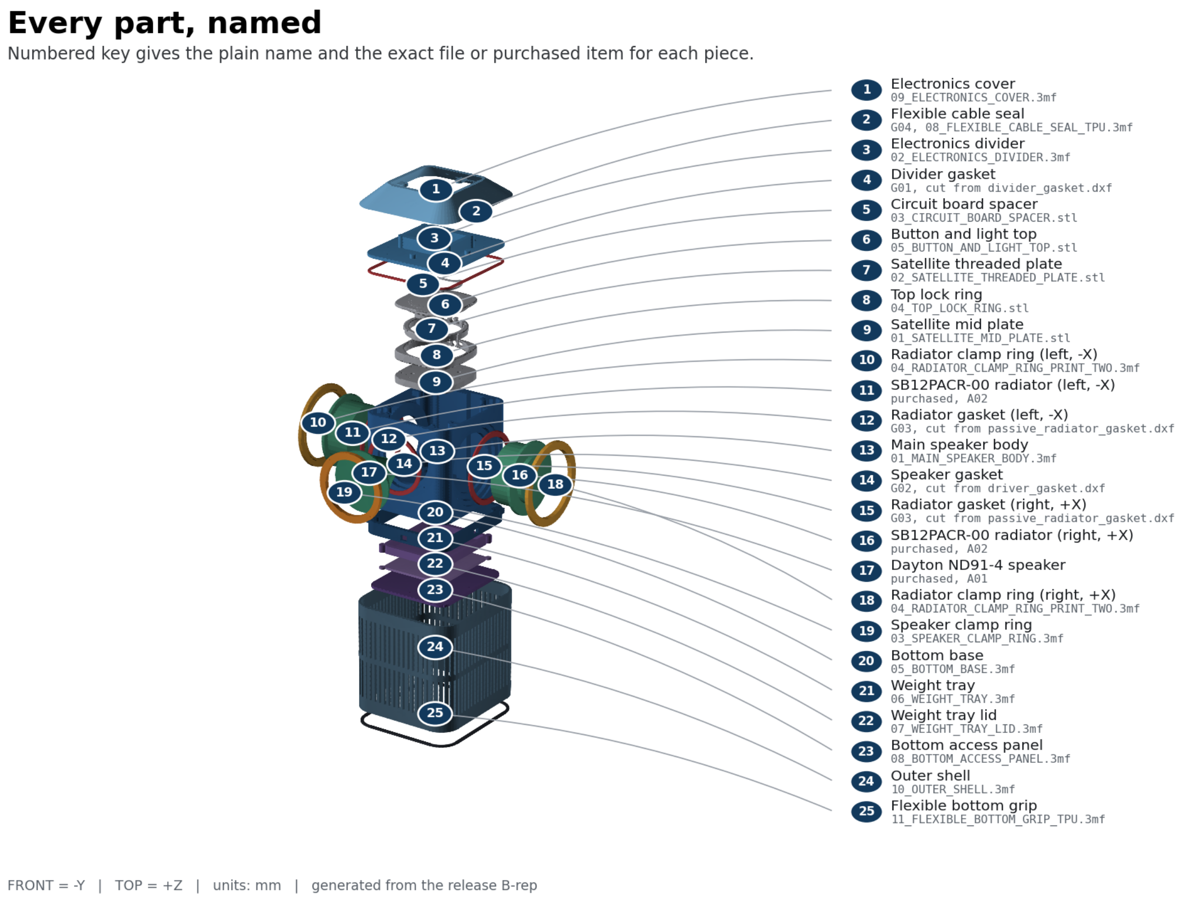
Each gasket must be one continuous piece. Do not splice, join, or stretch a gasket to fit.



Where each of the four seals sits

Phase 6 — Lay out and check every part

Put everything on a clean table and account for it before you start. The diagram below names every single piece and tells you which file it came from or which item you purchased.



Every part in the build, with its file name

Check that you have all of it:

• All 12 printed Ultra parts, with two radiator clamp rings.

• All 6 printed Satellite top parts.

• One ND91-4 speaker and two SB12PACR-00 radiators.

• Three cut foam gaskets and one printed TPU cable seal.

• Every screw and insert in `FASTENERS.csv`.

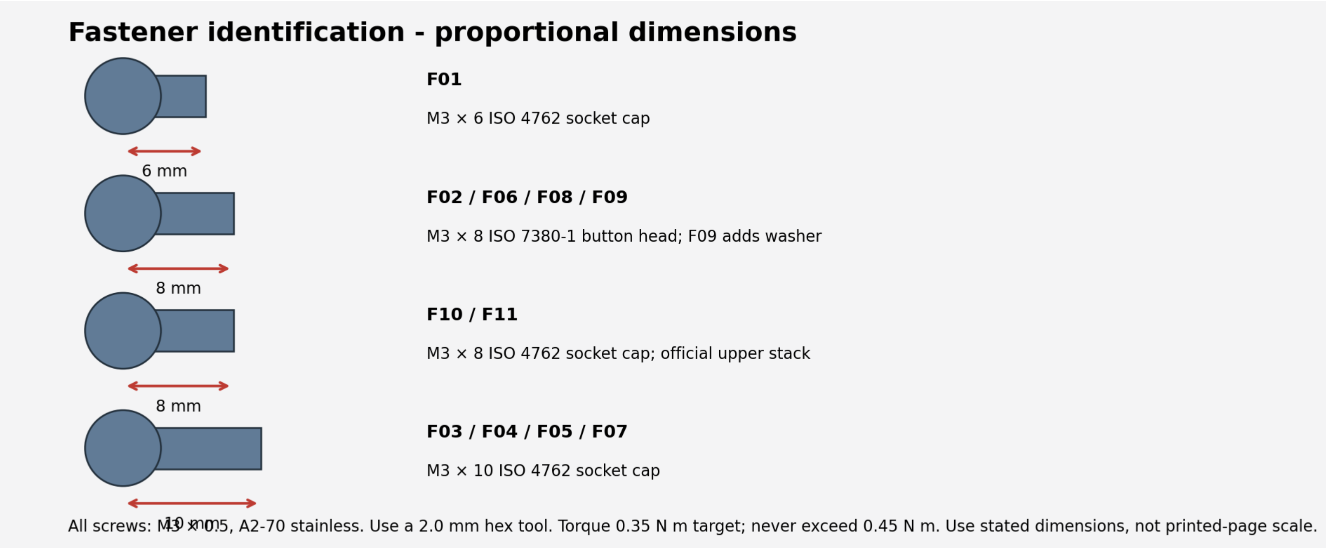
• Two steel ballast plates, deburred and dry.

• Two matched radiator tuning stacks, weighed to

0.78 g each.

• One red/black speaker lead with the correct plug.

The screws all look similar, so identify them by length before you begin:



Screw lengths and where each one is used

Do not begin assembly with a missing part.

Phase 7 — Assemble

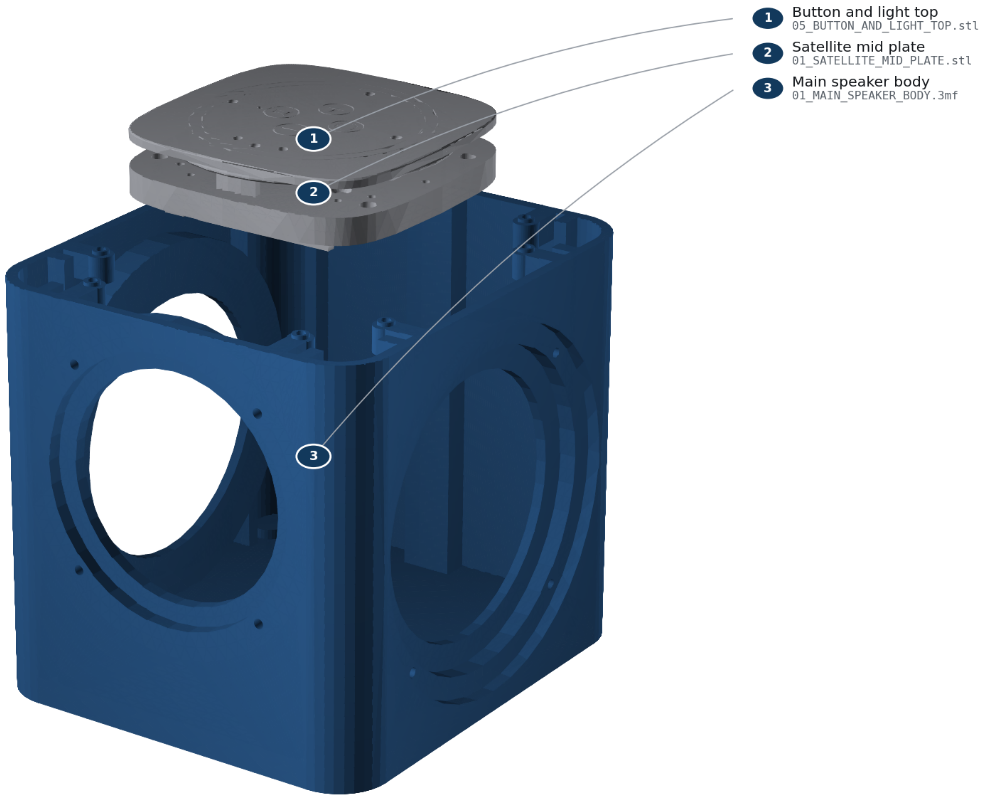
Nine steps, in this order. Each one shows what to install, which screws to use, which seal it captures, and how to tell it is right.

Tighten every screw into a printed part gently, using the short arm of the 2 mm hex key: 0.35 N m is the target and 0.45 N m is the absolute maximum. Stop as soon as the parts meet evenly. Do not keep turning "for luck".

Step 1 of 9 — Identify and inspect the hardware

STEP 1 **Identify and inspect the hardware**

Confirm Batch 1 hardware and inspect every printed sealing land.



Build step 1: Identify and inspect the hardware

You need: Batch 1 Core rev4.1 and HAT rev4.1; O01 official_mid_plate; O02 official_mid_plate_threads; O03 official_pcb_spacer; O04 official_lock_ring; O05 official_top_plate; O06 official_top_plate_snap_in_diffuser_ring

Screws: none **Seal:** none **Tools:** bright light; calipers

Do this: Confirm the board revision labels. Reject Batch 2 / Satellite1.1. Check off all six required official filenames in OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL and every required custom 3MF in the Printing Guide. Inspect every sealing face and remove strings without rounding an edge.

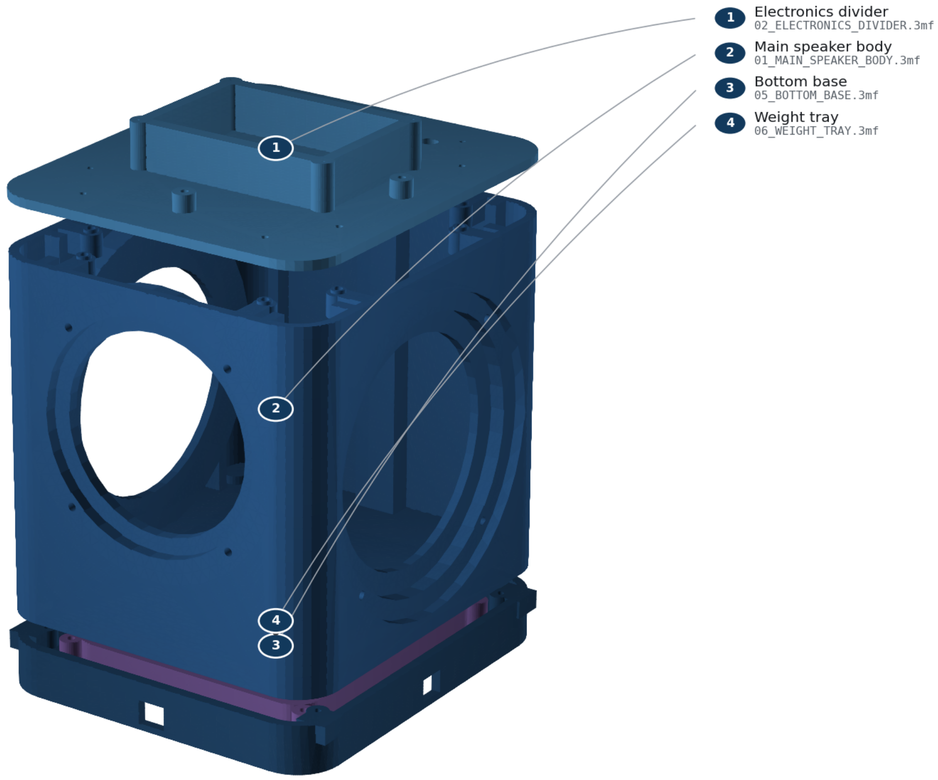
It is right when: Correct Batch 1 hardware and every required printed part are present; no crack, warp, blocked bore, or damaged gasket land.

> Careful: Do not force or approximately place the Core. Its exact stack placement requires the physical official hardware.

Step 2 of 9 — Install and cold-check all M3 inserts

STEP 2 **Install and cold-check all M3 inserts**

Install every insert square in its blind bore; let it cool before checking.



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Build step 2: Install and cold-check all M3 inserts

You need: main_cabinet, pressure_divider, base_skirt, ballast_cartridge, outer_shell

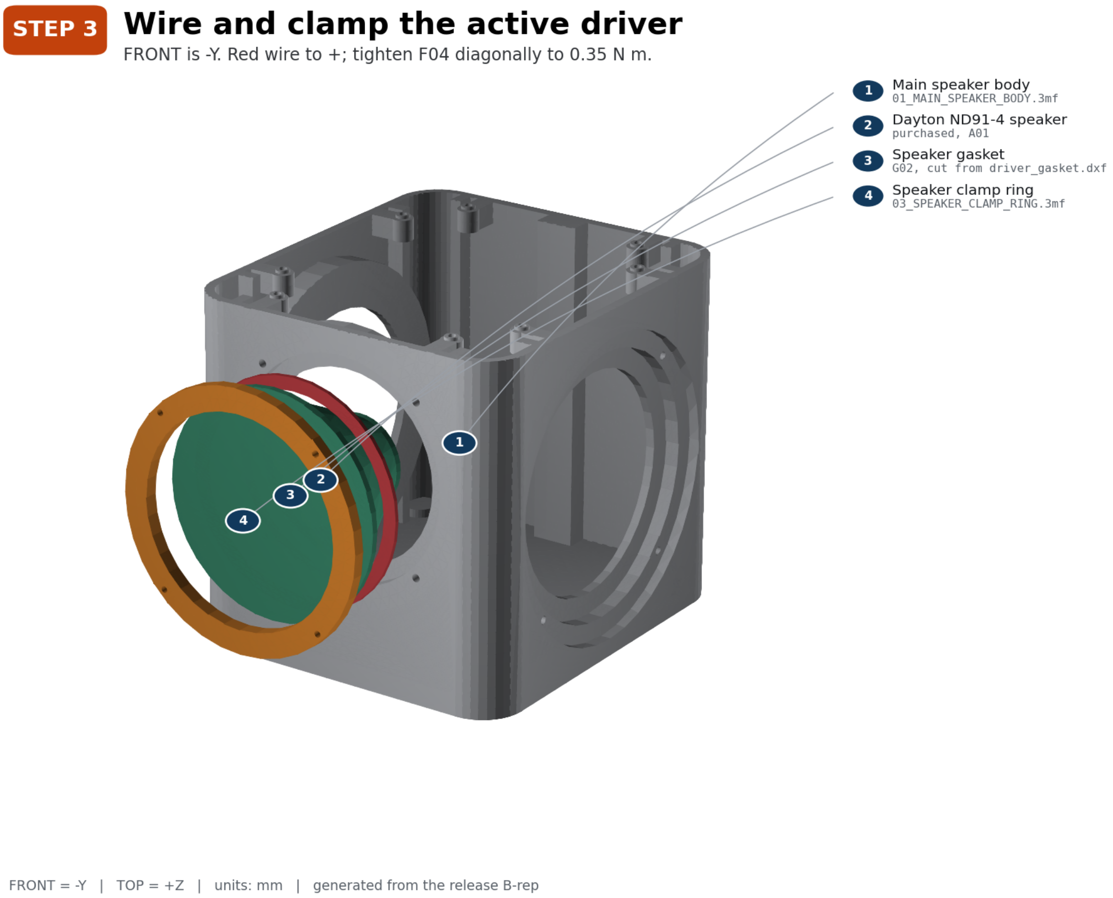
Screws: H01 inserts **Seal:** none **Tools:** temperature-controlled iron; M3 insert tip; square

Do this: At 250-270 C, press each insert into its labeled blind bore until flush and square. Let every insert cool for five minutes. Thread an M3 screw by hand for three turns.

It is right when: No insert spins, tilts, protrudes, or blocks before three turns.

> Careful: Do not torque a hot insert. Fumes and the iron can burn; ventilate and use eye protection.

Step 3 of 9 — Wire and clamp the active driver



Build step 3: Wire and clamp the active driver

You need: main_cabinet, Dayton ND91-4, driver_gasket, active_driver_clamp_ring, JST-XH lead

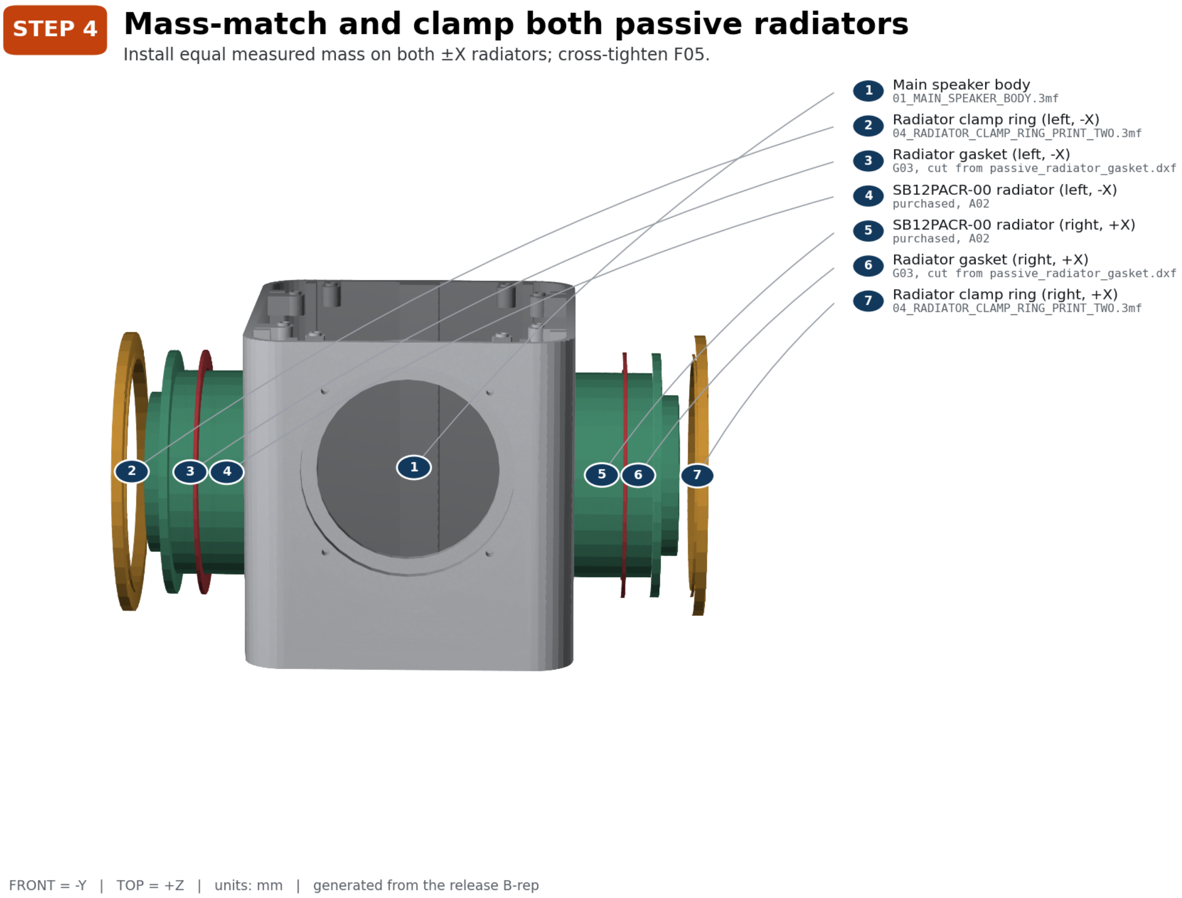
Screws: F04, 4 screws **Seal:** G02 **Tools:** 2.0 mm hex; crimper or soldering iron; polarity tester

Do this: Mark the red conductor positive. Connect red to the terminal marked + and black to -. Face the terminals upward. Center G02, seat the driver from the -Y/front side, fit the clamp ring, and tighten F04 in two diagonal passes to 0.35 N m; never exceed 0.45 N m.

It is right when: Ring bottoms evenly; gasket is not visible in the bore; cone moves outward on a brief 1.5 V positive polarity pulse.

> Careful: Use only a brief low-voltage polarity pulse. Never connect a loose driver to the powered HAT.

Step 4 of 9 — Mass-match and clamp both passive radiators



Build step 4: Mass-match and clamp both passive radiators

You need: 2 SB12PACR-00, 2 passive_radiator_gaskets, 2 clamp rings, matched M6 washer stacks

Screws: F05, 8 screws total **Seal:** G03, one per side **Tools:** 0.01 g scale; 2.0 mm hex

Do this: Weigh two identical tuning stacks to the value in reports/acoustics/summary.json. Secure one to each M6 post. Install radiators on +/-X with matching orientation, then tighten each F05 crosswise to 0.35 N m; never exceed 0.45 N m.

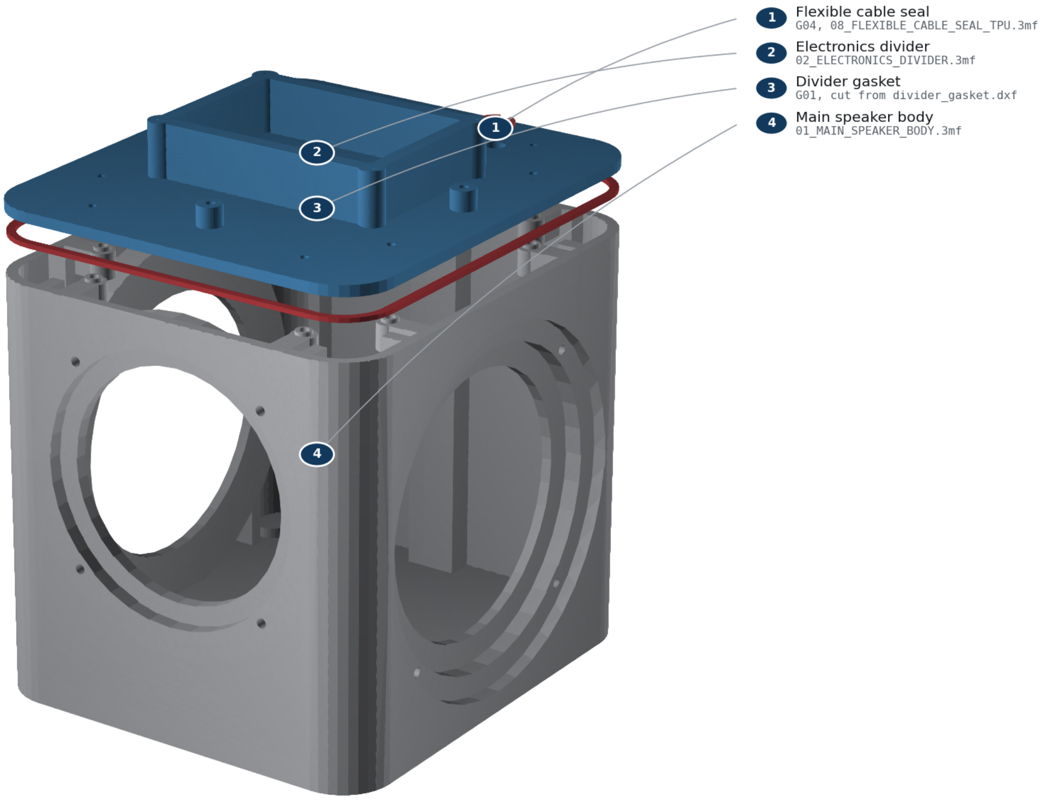
It is right when: Added masses match within 0.02 g; both rings bottom evenly; surrounds move freely and do not touch the shell keep-out.

> Careful: Unequal mass defeats reaction-force cancellation. Do not press on either cone.

Step 5 of 9 — Route the cable, close the divider, and leak-check

STEP 5 **Route the cable, close the divider, and leak-check**

Close G01, gross-screen at only 100–250 Pa, then fit G04 flange-up.



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Build step 5: Route the cable, close the divider, and leak-check

You need: pressure_divider, divider_gasket, leak_test_adapter, cable_gland

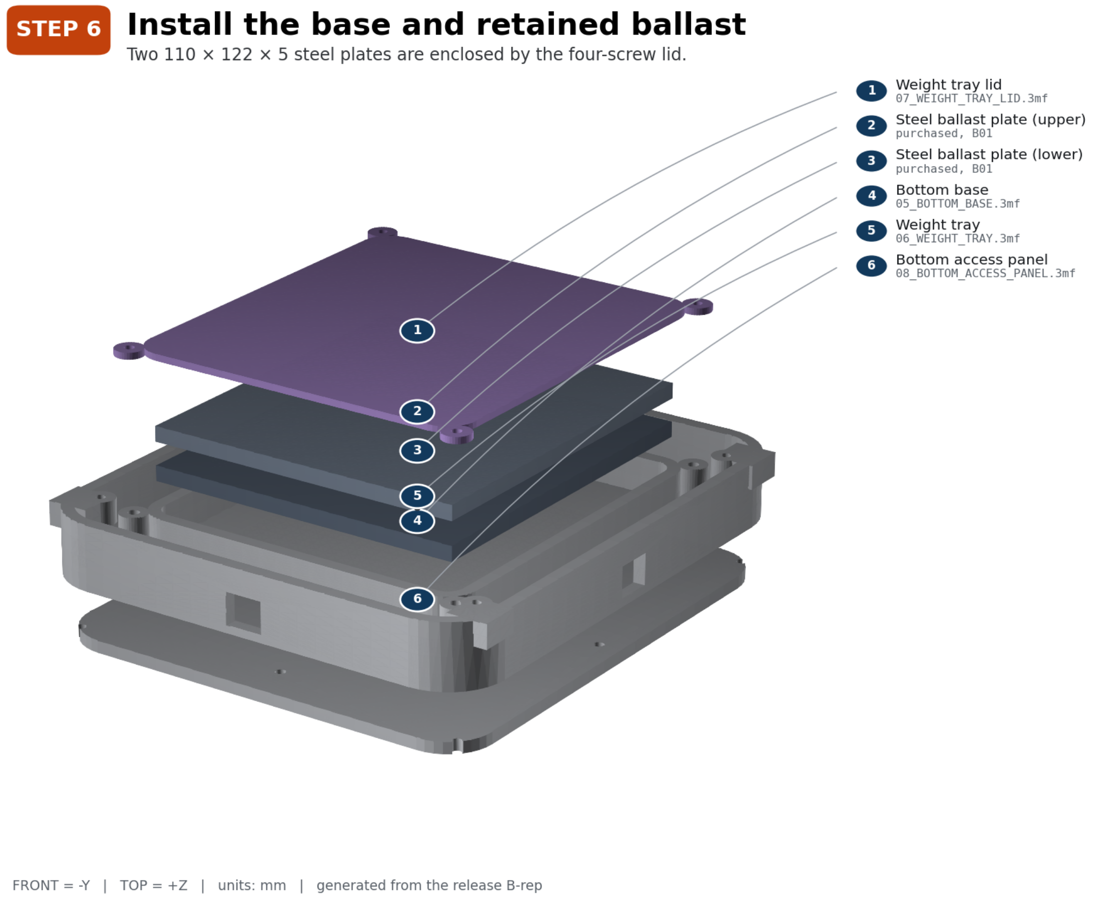
Screws: F03, 8 screws **Seal:** G01; temporary adapter then G04 **Tools:** 2.0 mm hex; hand bulb; 0-500 Pa gauge; leak-detection solution

Do this: Pass both conductors through the divider. Fit the temporary adapter over them, place G01 without twists, and tighten F03 in a star pattern to 0.35 N m. Apply only 100-250 Pa with a hand bulb. Brush leak solution on external gasket seams; no bubbles are allowed. Vent, pull the adapter upward, and install G04 with its flange toward the electronics bay.

It is right when: No growing bubbles, abnormal diaphragm displacement, or audible leak; final gland cannot rotate or lift by finger force.

> Careful: Never use shop air, never exceed 250 Pa, and keep liquid away from electronics. This is a gross-leak screen, not an acoustic-Q measurement.

Step 6 of 9 — Install the base and retained ballast



Build step 6: Install the base and retained ballast

You need: base_skirt, ballast_cartridge, 2 steel plates, ballast lid, bottom_service_plate

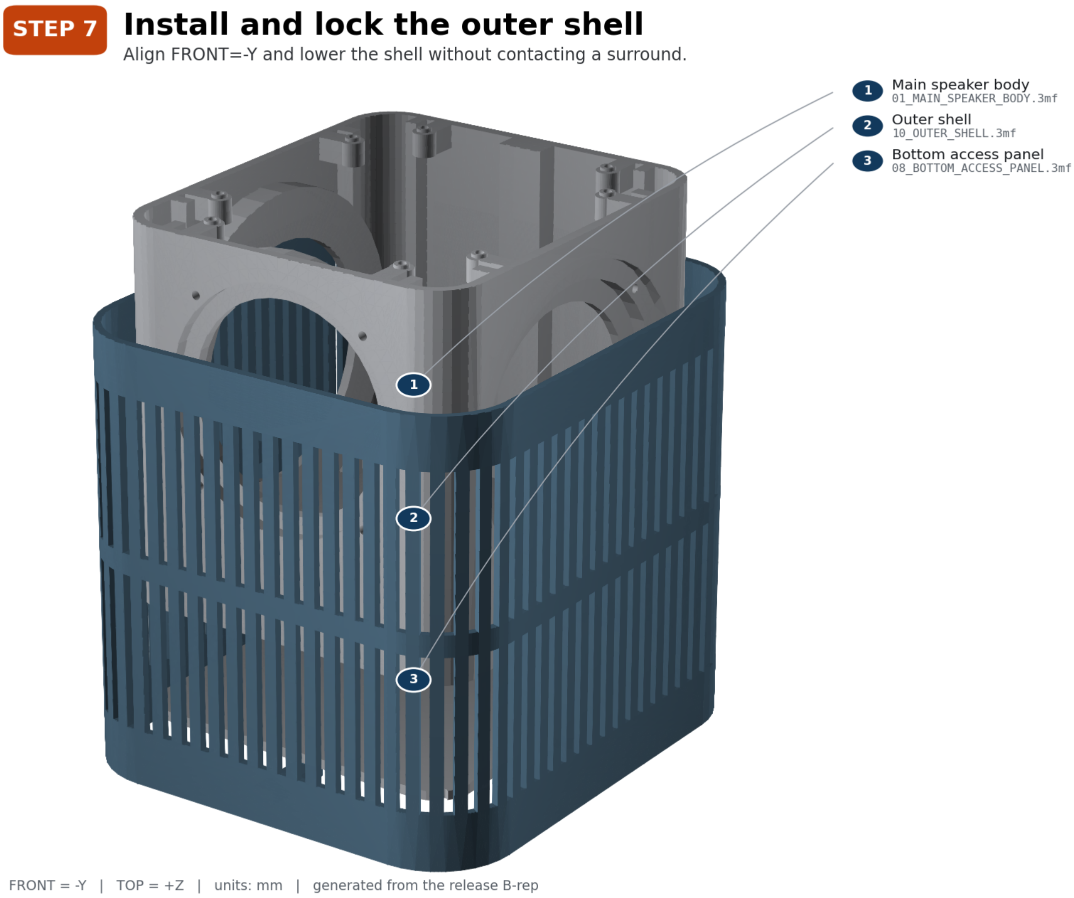
Screws: F06, F07, F08 **Seal:** none **Tools:** 2.0 mm hex; scale

Do this: Attach the base skirt with F07. Place both deburred dry plates flat in the cartridge; there must be no rocking. Install the lid with F06, insert the cartridge from below, and capture it with the bottom service plate using F08.

It is right when: Cartridge mass is approximately 1054 g; no plate moves when shaken gently; all four lid screws engage at least 3 mm.

> Careful: The steel stack is heavy. Keep fingers clear and do not operate the unit without the retained lid and service plate.

Step 7 of 9 — Install and lock the outer shell



Build step 7: Install and lock the outer shell

You need: outer_shell; lower assembly

Screws: F09, 4 screws with nylon washers **Seal:** none **Tools:** 2.0 mm hex

Do this: Align FRONT with -Y and slide the shell downward without touching either surround. Invert on a soft mat and install F09 through the bottom service plate into the shell bosses.

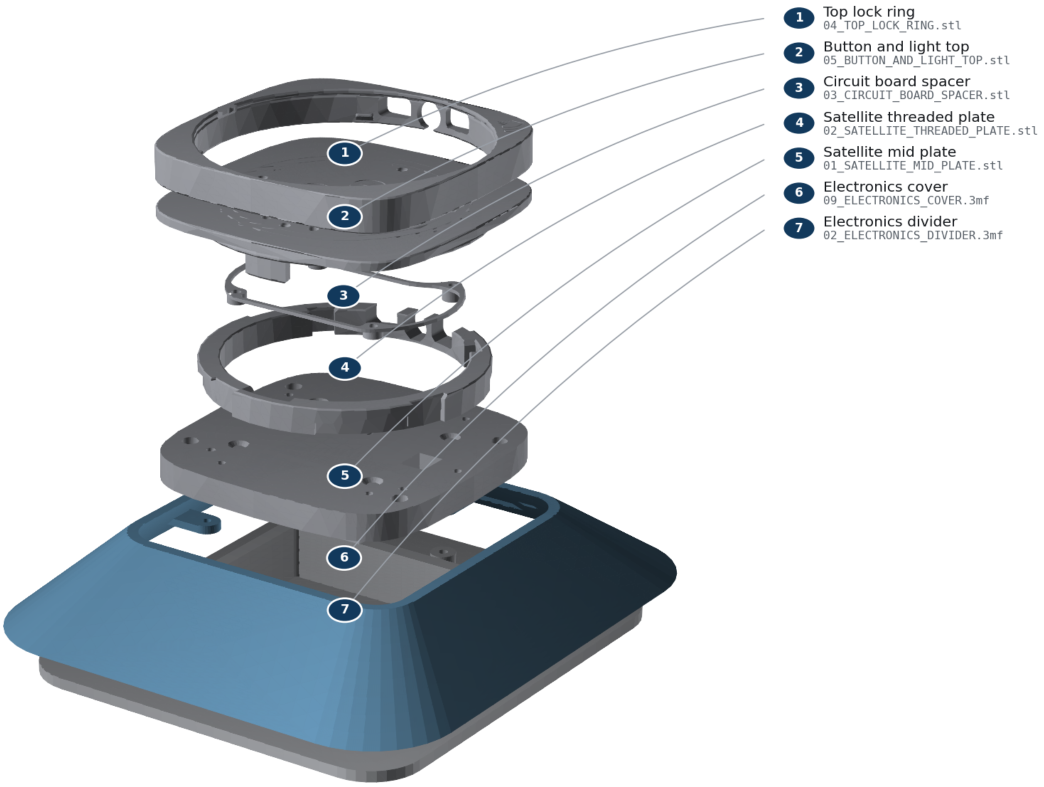
It is right when: Every slot is clear; shell has an even reveal and at least 2 mm moving-part clearance; no wire is visible near a cone.

> Careful: Stop if the shell contacts a clamp ring or surround. Do not flex the shell over an obstruction.

Step 8 of 9 — Install the shroud and official Batch 1 upper stack

STEP 8 **Install the shroud and official Batch 1 upper stack**

Mount the official mid-plate to the measured four-point interface; preserve USB-C.



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Build step 8: Install the shroud and official Batch 1 upper stack

You need: electronics_shroud; O01-O06 official prints; Batch 1 HAT/Core

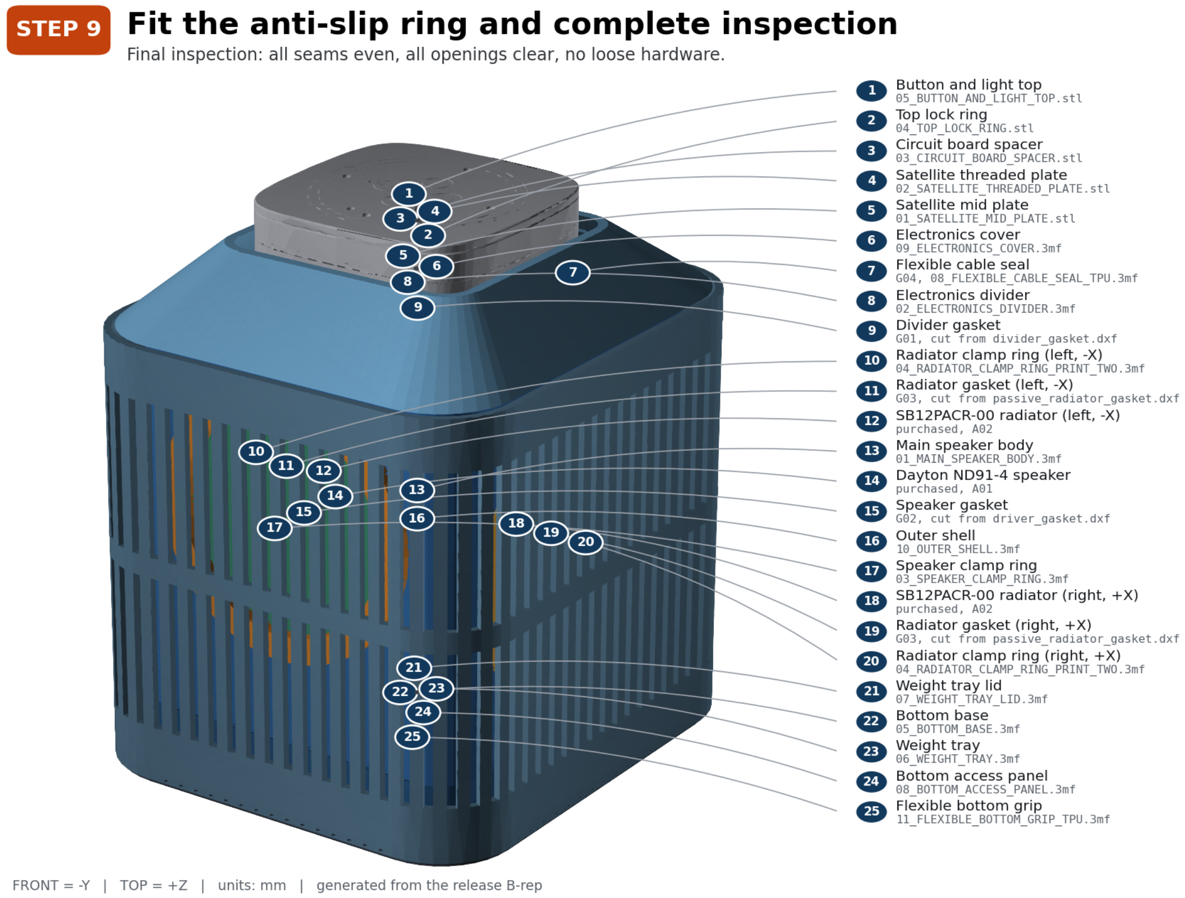
Screws: F01, F02, F10, and F11; 4 of each **Seal:** none; electronics bay is outside the acoustic chamber **Tools:** 2.0 mm hex; ESD-safe bench

Do this: Bolt the shroud to its four outboard bosses with F02. Seat O01 on the four measured divider bosses and install F01. Snap O06 into O05 (or use both O07/O08 during a multi-material O05 print; never install O06 and O08 together). Align O03's taller standoffs with the I/O side and locate the HAT. Install the Core/HAT using the official Batch 1 sequence. Align the logos and I/O on O04/O05, engage the snaps, and rotate the lock ring. Align O02's four nubs with O01 and keep I/O toward rear/+Y. Connect the keyed JST-XH speaker plug before closure.

It is right when: Mid-plate sits on all four bosses; USB-C remains reachable; cable has service slack and cannot enter a moving-part envelope; buttons click and diffuser/LED apertures remain clear.

> Careful: Core placement is REQUIRES_PHYSICAL_VALIDATION. Follow the official Batch 1 instructions and stop at any collision; do not improvise a transform from the CAD envelope.

Step 9 of 9 — Fit the anti-slip ring and complete inspection



Build step 9: Fit the anti-slip ring and complete inspection

You need: anti_slip_ring; complete assembly

Screws: none **Seal:** inspect G01-G04 **Tools:** hands; flashlight

Do this: Stretch the TPU ring evenly around the bottom rim. Set the unit upright and inspect all seams, fastener heads, slots, cable exits, buttons, and moving components.

It is right when: Unit stands without rocking; no rattle is heard during gentle handling; every fastener is present and every seal is continuously compressed.

> Careful: Do not power the unit until the commissioning checklist is ready.

Phase 8 — Test before you use it

Complete every check in `TESTING_AND_COMMISSIONING_GUIDE.pdf`. The short version:

• The gentle 100-250 Pa leak screen shows no growing bubble at any seal.

• The speaker cone moves outward on a brief positive polarity pulse.

• Both radiators move freely and never scrape.

• Every button clicks once and returns.

• Every LED segment lights evenly.

• All four microphones work.

• USB-C plugs in and out without touching the shell.

• Wi-Fi connects, and you have recorded the signal beside a bare kit.

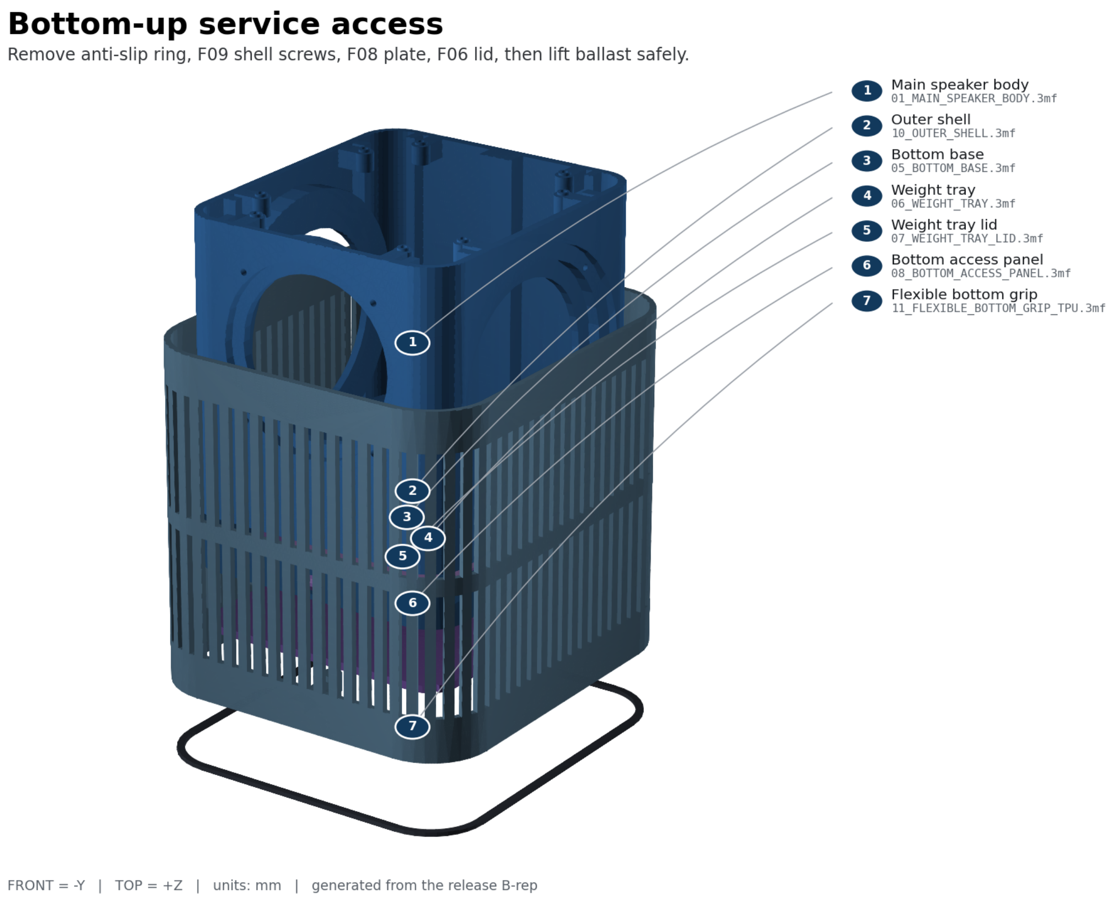
• There is no buzz, rattle, air whistle, overheating, or softened plastic.

Never use shop air on this enclosure, and never exceed 250 Pa.

Stop using the unit if any check fails. Fix the cause and repeat the test.

Phase 9 — Opening it again later

Disconnect power and wait five minutes. `MAINTENANCE_GUIDE.pdf` gives the safe removal order for each subassembly.



The order things come apart in

Replace any gasket you disturb. Never cut a wire to get a part out, and never reuse a torn or permanently flattened seal.

If something goes wrong

Symptom	Most likely cause	What to do
Calibration piece is warped	chamber too cool, or bed not clean	fix the printer and reprint; never compensate for a warped part
Every hole is too small	over-extrusion or elephant-foot compensation	fix flow and first-layer compensation before adding a CAD offset
Insert cracks the boss	bore too small or iron too hot	choose a larger bore or reduce dwell; never add torque

Symptom	Most likely cause	What to do
Speaker will not sit flat	print strings, or cutout too small	remove strings only; correct the cutout and reprint the coupon
Cable seal leaks or spins	conductor too thick, or wet TPU	confirm conductor OD is 1.8 mm or less, dry the TPU, reprint
Shell will not slide on	clamp ring or surround contact	stop immediately; find the contact rather than forcing the shell
Bubbles during the leak check	gasket twisted, pinched, or a screw left loose	vent, reopen, replace the gasket, and reseat evenly
One radiator moves more than the other	unequal tuning mass	reweigh both stacks; they must match within 0.02 g

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What is still unproven

The geometry, exports, schedules, and documents in this release pass their digital gates. No completed physical unit has been measured. That means your own calibration, leak, acoustic, thermal, Wi-Fi, and microphone checks are the first real evidence this design works. Record what you find.

`ENGINEERING_APPENDIX.md` lists every open risk, its consequence, and how to close it.

Source commit c83cac321c50. No physical validation has been performed.